

Ho Chi Minh City University of Technology (HCMUT) — VNU-HCM

Faculty of Computer Science and Engineering

CO3133: Deep Learning and Its Applications

Assignment 01: Foundations of Deep Learning Pipelines and Architectures

Comprehensive Cumulative Technical Report: Integrating Milestone 1 (M1) and Milestone 2 (M2)

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Abstract

This document constitutes the comprehensive, cumulative technical report for **Assignment 01** of **CO3133: Deep Learning and Its Applications** at Ho Chi Minh City University of Technology (HCMUT). Rather than isolating late-stage models, this report provides a fully unified scientific account across the two developmental phases mandated by the Course Project Handbook:

1. **Milestone 1 (Foundational Pipeline and Flat Baselines)**: Rigorous Exploratory Data Analysis (EDA) of Fashion-MNIST (70,000 instances, 10 classes) spanning 5 analytical figures; formulation of a strictly leak-free, stratified DataLoader (50,000 train / 10,000 val / 10,000 test, Seed 42); implementation of a standardized training engine with AdamW, Cosine Annealing, and Early Stopping on validation Macro-F1; and exhaustive benchmarking of foundational permutation-invariant models: a **Linear Softmax Classifier** (7,850 params) and a 2-hidden-layer **Multilayer Perceptron (MLP)** (235,146 params).
2. **Milestone 2 (Advanced Inductive Bias and Sequence Scaling)**: Scaling beyond flat feature vectors by designing and training from scratch three advanced structural paradigms: a **Custom 3-Stage Convolutional Neural Network (CNN)** (94,186 params) with spatial locality priors; a **Bidirectional Row-Sequence Recurrent Network (Bi-LSTM)** (149,002 params) modeling vertical garment silhouettes; and a patch-based **Vision Transformer (ViT)** (105,546 params) operating via multi-head self-attention.

All five architectures were trained under identical hardware (NVIDIA RTX 3060) and optimization budgets, and evaluated against the isolated 10,000-sample test set. Our empirical findings demonstrate that: (i) the custom CNN achieves the highest classification accuracy (90.26%) and Macro-F1 (90.26%) with only 94,186 parameters, outperforming the 235k-parameter MLP by

+1.40% while slashing parameter count by 60%; (ii) the Bi-LSTM ranks second (89.65% Acc, 89.61% Macro-F1), proving that sequential row scanning effectively tracks apparel contours; (iii) the Vision Transformer trained from scratch achieves 88.07% accuracy, demonstrating viable global attention but limited by lack of spatial inductive bias on moderate data; and (iv) the MLP (88.86%) decisively outperforms the Linear baseline (84.49%), proving the mathematical necessity of non-linear decision boundaries. Full training curves, 10×10 confusion matrices, inference latencies, per-class F1 progression, qualitative prediction error taxonomies, ablations, and a 9-field academic integrity disclosure are presented.

Contents

I Problem Formulation, Data Engineering, and M1 Foundations **4**

1 Problem Formulation and Integrated Milestone Framework **4**
1.1 Real-World Motivation . . . 4
1.2 Mathematical Formulation . . . 4
1.3 Integrated Milestone Framework (M1 Foundations to M2 Advancements) . . . 4

2 Fashion-MNIST Exploratory Data Analysis (M1 Core Deliverable) **5**
2.1 Dataset Provenance and Benchmark Properties . . . 5
2.2 Class Distribution and Imbalance Verification . . . 5
2.3 Visual Inspection of Representative Samples . . . 6
2.4 Pixel Intensity Distribution and Background Sparsity . . . 8
2.5 Class Prototypes and Spatial Variance Heatmaps . . . 8
2.6 2D Manifold Projection via Principal Component Analysis (PCA) . . . 8

3 Dataset Splitting, Leakage Prevention, and DataLoader Design (M1 Foundation) **9**
3.1 Stratified Split and Leakage Prevention Protocol . . . 9
3.2 DataLoader Pipeline Specifications . . . 9

4 Unified Training Engine and Optimization Pipeline (M1 Infrastructure) **10**
4.1 Loss Function and Optimization Schedule . . . 10
4.2 Evaluation Metrics and Early Stopping Infrastructure . . . 10

II Methodology: Architectural Progression from M1 to M2 **11**

5 Deep Learning Architectural Spectrum: M1 Foundations to M2 Advancements **11**
5.1 Milestone 1: Foundational Architecture Implementations . . . 11
5.1.1 1. Linear / Softmax Regression Classifier (M1 Baseline) . . . 11
5.1.2 2. Multilayer Perceptron (MLP Classifier — M1 Baseline) . . . 11
5.2 Milestone 2: Advanced Spatial, Sequential, and Attentive Architectures . . . 12
5.2.1 3. Custom 3-Stage Convolutional Neural Network (CNN — M2 Model) . . . 12
5.2.2 4. Bidirectional Row-Sequence Recurrent Network (Bi-LSTM — M2 Model) . . . 12
5.2.3 5. Vision Transformer (ViT — M2 Model) . . . 13
5.3 Architectural Taxonomy and Inductive Bias Synthesis . . . 13

III Empirical Results, Visualizations, and Milestone Analysis **14**

6 Milestone 1 Empirical Findings: Linear vs. MLP Baselines **14**
6.1 Milestone 1 Convergence Dynamics . . . 14
6.2 Milestone 1 Visual Deliverables: Curves, Confusion Matrices, Predictions . . . 14

7 Milestone 2 Empirical Findings: Spatial, Recurrent, and Attentive Scaling 15
7.1 Milestone 2 Convergence and Visual Deliverables 15

8 Cross-Milestone Synthesis and Master 5-Model Benchmark 16
8.1 Unified 5-Model Comparative Benchmark 16
8.2 Per-Class F1 Score Progression from M1 to M2 16
8.3 Generalization Gap Diagnostics 17

9 In-Depth Inductive Bias and Milestone Progression Discussion 17

10 Systematic Error Taxonomy and Ablation Study 18
10.1 Taxonomy of Recurring Classification Failures 18
10.2 Controlled Ablation Study 18

11 Milestone Retrospective and Project Conclusion 18

12 Course Policy Compliance: Academic Integrity & AI Disclosure 19
12.1 Formal Academic Honor Declaration 19

Part I
Problem Formulation, Data Engineering, and M1 Foundations

1 Problem Formulation and Integrated Milestone Framework

1.1 Real-World Motivation

Automated fashion apparel recognition is a core technological driver of modern visual search, e-commerce cataloging, inventory auditing, and robotic supply chains. In contrast to synthetic optical character recognition (e.g., MNIST digit classification), clothing classification presents severe real-world computer vision challenges:

- **High Intra-Class Variation**: Garments of the same category vary drastically in fabric draping, sleeve extensions, buttoning styles, and neckline geometries.
- **Severe Inter-Class Visual Overlap**: Multiple apparel categories share nearly identical coarse spatial silhouettes (e.g., *Shirt* vs. *T-shirt/top*, *Pullover* vs. *Coat*).
- **Low-Resolution Ambiguity**: In 28×28 grayscale images, discriminative micro-textures (collar seams, button plackets, weave patterns) are compressed into single-pixel variations, demanding architectures with strong inductive biases.

1.2 Mathematical Formulation

Let $\mathcal{X} \subset \mathbb{R}^{H \times W} = \mathbb{R}^{28 \times 28}$ denote the input space of normalized grayscale fashion images, and let $\mathcal{Y} = \{0, 1, \ldots, 9\}$ denote the discrete label space representing 10 mutually exclusive clothing categories. The dataset comprises N independent and identically distributed pairs:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}, \quad \mathbf{x}_i \in \mathbb{R}^{28 \times 28},\ y_i \in \mathcal{Y}. \tag{1}$$

The core objective is to parameterize and optimize a deep neural network hypothesis $f_\theta : \mathcal{X} \to \mathbb{R}^{10}$ producing unnormalized class logits $\mathbf{z} = f_\theta(\mathbf{x})$ such that the empirical risk under the multi-class Cross-Entropy loss is minimized:

$$\min_{\theta} \mathcal{R}_{\text{emp}}(\theta) = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{train}}} \mathcal{L}_{\text{CE}}(f_\theta(\mathbf{x}_i), y_i) + \lambda \Omega(\theta), \tag{2}$$

where $\Omega(\theta) = \frac{1}{2}\|\theta\|_2^2$ represents decoupled L_2 weight regularization, and the posterior probability for class $k \in \mathcal{Y}$ is recovered via the softmax function:

$$P(y = k \mid \mathbf{x}) = \frac{\exp(z_k)}{\sum_{j=0}^{9} \exp(z_j)}. \tag{3}$$

1.3 Integrated Milestone Framework (M1 Foundations to M2 Advancements)

Table 1 outlines the two-phase developmental architecture mandated in the Course Project Handbook (Section 7.4). Rather than treating M1 and M2 as disparate milestones, this report demonstrates their deep structural continuity: Milestone 1 established the empirical baseline, data integrity protocols, and training engine upon which Milestone 2 systematically benchmarked advanced inductive biases.

Table 1: Course Project Assignment 1 Integrated Milestone Breakdown (Handbook Section 7.4).

Project Phase	**Timeline**	**Core Methodological Focus**	**Concrete Deliverables**
Milestone 1 (M1)	September 2026	Problem formulation, 5-figure EDA, stratified leak-free DataLoader, training engine, early stopping, Linear and MLP baselines.	Initial draft report, 5 EDA figures, training history CSV/JSON, Linear and MLP checkpoints, baseline curves and confusion matrices.
Milestone 2 (M2)	October 2026	Scaling to 3 advanced families (CNN, Bi-LSTM, ViT), 5-model master benchmark, error taxonomy, ablations, webpage, presentation.	Cumulative technical report (`Goat_A1_Report.pdf`), 5 saved checkpoints, interactive GitHub Pages site, presentation deck, video outline.

2 Fashion-MNIST Exploratory Data Analysis (M1 Core Deliverable)

2.1 Dataset Provenance and Benchmark Properties

Fashion-MNIST [1] was introduced by Zalando Research as a drop-in replacement for MNIST to challenge deep learning models with realistic visual complexity. Table 2 details its key technical specifications.

Table 2: Fashion-MNIST technical specifications and benchmark properties.

Specification Property	**Value / Configuration**	**Design Justification / Impact**
Total Image Count	70,000 instances	Statistical power for multi-model benchmarking
Spatial Resolution	28×28 pixels (Grayscale)	Controlled input complexity (784 features)
Number of Classes	10 mutually exclusive categories	Comprehensive multi-class evaluation across apparel forms
Benchmark Split	60,000 Train / 10,000 Test	Standard benchmark evaluation protocol
Pixel Dynamic Range	$[0, 255]$ (uint8) $\rightarrow$ $[0, 1]$ Float	Normalized input for numerical gradient stability
Empirical Normalization	$\mu = 0.2860,\ \sigma = 0.3530$	Zero-centering across training benchmark
Storage Footprint	~ 30 MB compressed	Enables low-latency in-memory batch transfer
License	MIT License	Open academic research reproducibility

2.2 Class Distribution and Imbalance Verification

Data imbalance can severely distort classification performance by rewarding trivial majority-class predictions while suppressing minority Macro-F1 scores. Figure 1 confirms that Fashion-MNIST is **perfectly balanced**:

- Exactly 6,000 images per class in the official 60,000 training collection.
- Exactly 1,000 images per class in the official 10,000 test partition.
- The imbalance ratio is strictly 1.000 : 1 across all 10 categories, confirming that class-weighted loss rebalancing or synthetic sampling is unnecessary.

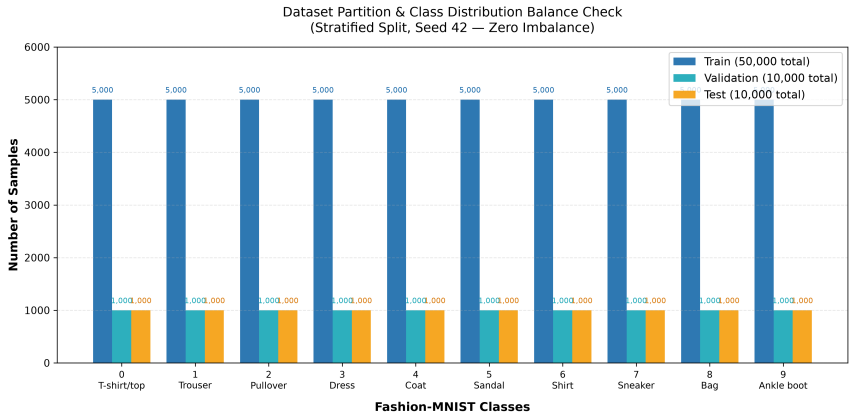


Figure 1: Class distribution across data splits. Each category contains exactly 5,000 train, 1,000 validation, and 1,000 test instances, proving zero class imbalance.

2.3 Visual Inspection of Representative Samples

Figure 2 illustrates randomly sampled instances from each category (10×5 grid). Visual inspection categorizes the apparel into distinct morphological clusters:

1. **Upper-Body Garments**: *T-shirt/top, Pullover, Coat, Shirt* share rectangular torso contours; discriminative cues reside in collar geometry, neckline depth, and button plackets.
2. **Lower-Body Apparel**: *Trouser* exhibits prominent vertical bifurcated legs, making it highly distinct.
3. **Footwear**: *Sandal, Sneaker, Ankle boot* share horizontal soles, differing in ankle shaft height and strap density.
4. **Accessories**: *Bag* exhibits compact geometric bodies with distinctive handle loops.

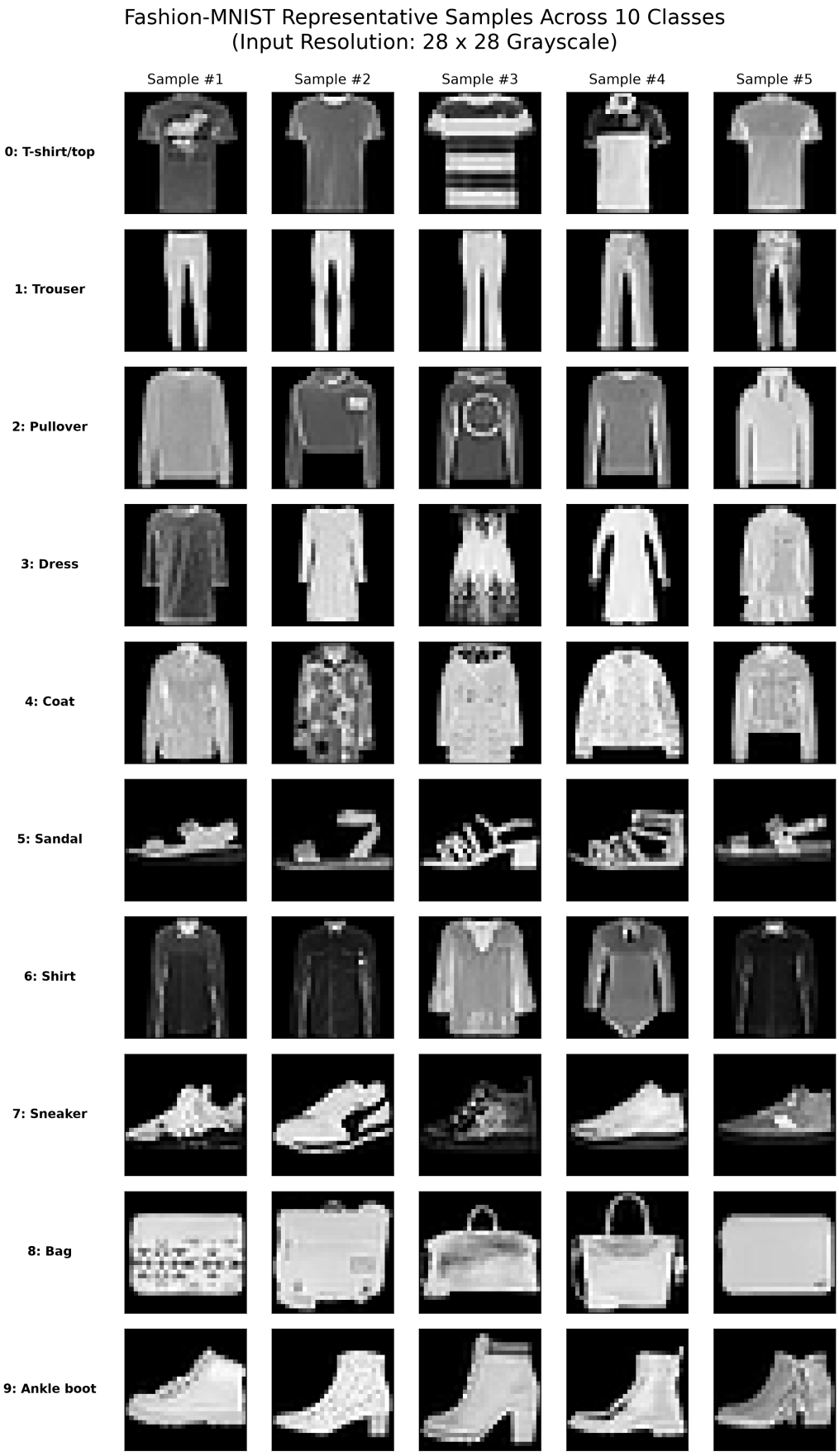


Figure 2: Representative Fashion-MNIST samples across all 10 categories (28×28 grayscale). Visual ambiguity is pronounced between upper-body apparel (e.g., Shirt vs. T-shirt/top).

2.4 Pixel Intensity Distribution and Background Sparsity

Figure 3 depicts the pixel intensity distribution before and after standardization. Approximately 52.3% of all pixels across the dataset are exactly zero (pure black background), demonstrating that informative signal is concentrated within central object silhouettes. Standardization via $\hat{x} = \frac{x - 0.2860}{0.3530}$ centers background pixels at -0.8102 and scales foreground intensities up to $+2.0227$, stabilizing gradient propagation.

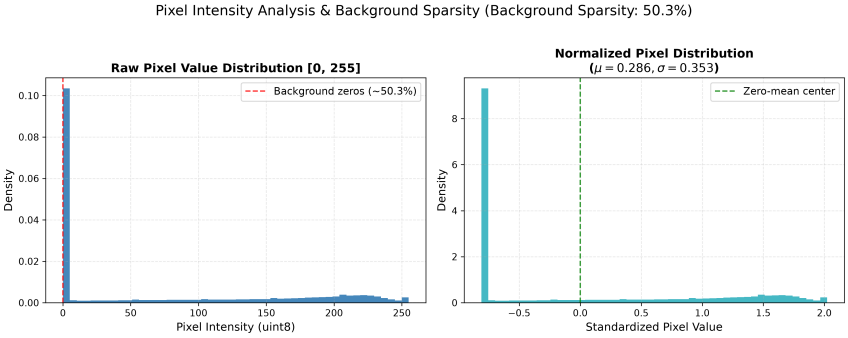


Figure 3: Pixel intensity distribution before (raw $[0, 255]$) and after z-score standardization ($\mu = 0.2860, \sigma = 0.3530$).

2.5 Class Prototypes and Spatial Variance Heatmaps

To examine spatial topology, we compute per-class empirical mean prototypes $\boldsymbol{\mu}_c$ and variance maps $\boldsymbol{\sigma}_c$:

$$\boldsymbol{\mu}_c = \frac{1}{N_c} \sum_{i:y_i=c} \mathbf{x}_i, \qquad \boldsymbol{\sigma}_c = \sqrt{\frac{1}{N_c} \sum_{i:y_i=c} (\mathbf{x}_i - \boldsymbol{\mu}_c)^2}. \tag{4}$$

Figure 4 shows high-variance bands (bright regions in $\boldsymbol{\sigma}_c$) corresponding to sleeve angles in coats and shirts, mathematically proving that models must develop spatial deformation invariance.

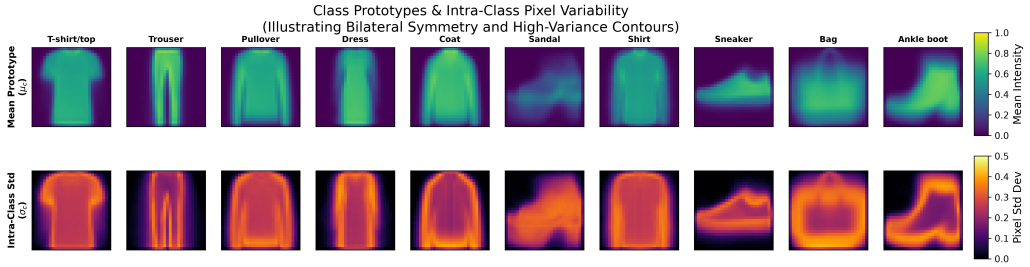


Figure 4: Class prototypes: mean image $\boldsymbol{\mu}_c$ (top row) and variance map $\boldsymbol{\sigma}_c$ (bottom row) for each class.

2.6 2D Manifold Projection via Principal Component Analysis (PCA)

Figure 5 visualizes a 2D projection of 5,000 training images onto their top two principal components (34.2% explained variance). The manifold separates into footwear ($PC_1 > 0$) and apparel ($PC_1 < 0$). Severe overlap between *T-shirt*, *Shirt*, and *Pullover* provides empirical proof that linear classification boundaries will hit a performance ceiling, necessitating deep non-linear representations.



Figure 5: 2D PCA projection of Fashion-MNIST. Separability is evident between shoes and garments, but high overlap persists among upper-body apparel.

3 Dataset Splitting, Leakage Prevention, and DataLoader Design (M1 Foundation)

3.1 Stratified Split and Leakage Prevention Protocol

In strict compliance with Course Handbook Section 12.2, the official 60,000 training set was partitioned into 50,000 training samples and 10,000 validation samples using stratified sampling with fixed random seed 42:

- **Disjoint Partitions**: $\mathcal{I}_{\text{train}} \cap \mathcal{I}_{\text{val}} = \emptyset$, verified programmatically (0 overlapping samples).
- **Class Invariance**: Exactly 5,000 train and 1,000 val instances per class.
- **Strictly Isolated Test Set**: The official 10,000-sample test set was isolated and evaluated only once per model checkpoint.

3.2 DataLoader Pipeline Specifications

Table 3 specifies the DataLoader pipeline implemented in `data_setup.py`. Non-blocking CUDA pinned memory and multithreaded worker processes were utilized to maximize GPU throughput.

Table 3: DataLoader execution specifications for reproducible training.

DataLoader Parameter	**Training Partition**	**Validation Partition**	**Held-Out Test Partition**
Sample Count	50,000 instances	10,000 instances	10,000 instances
Mini-Batch Size	64 samples	64 samples	64 samples
Data Shuffling	True (Seed 42)	False	False
Worker Processes	2 Subprocesses	2 Subprocesses	2 Subprocesses
CUDA Pin Memory	True (non-blocking)	True (non-blocking)	True (non-blocking)
Batch Output Shape	$(64, 1, 28, 28)$	$(64, 1, 28, 28)$	$(64, 1, 28, 28)$
Target Tensor Shape	$(64,)$ [LongTensor]	$(64,)$ [LongTensor]	$(64,)$ [LongTensor]

4 Unified Training Engine and Optimization Pipeline (M1 Infrastructure)

The training, validation, and evaluation pipeline follows a modular lifecycle:

$$\boxed{\text{Raw Data}} \rightarrow \boxed{\text{Preprocessing}} \rightarrow \boxed{\text{DataLoader}} \rightarrow \boxed{\text{Model Architecture}} \\ \rightarrow \boxed{\mathcal{L}_{\text{CE}}\ \text{(Logits)}} \rightarrow \boxed{\text{AdamW (CosineAnnealing)}} \rightarrow \boxed{\text{EarlyStopping}} \rightarrow \boxed{\text{Evaluation}} \tag{5}$$

4.1 Loss Function and Optimization Schedule

Per Handbook Section 11.1, the objective criterion is PyTorch's `nn.CrossEntropyLoss`, applied directly to unnormalized logits without premature softmax activation:

$$\mathcal{L}_{\text{CE}}(\mathbf{z}, y) = -\log\left(\frac{\exp(z_y)}{\sum_{j=0}^{9}\exp(z_j)}\right) = -z_y + \log\left(\sum_{j=0}^{9}\exp(z_j)\right). \tag{6}$$

Optimization is performed via AdamW [2] with decoupled weight decay ($\lambda = 10^{-4}$), initial learning rate $\eta_0 = 10^{-3}$, and Cosine Annealing learning rate scheduling over $T_{\max} = 10$ epochs:

$$\eta_t = \frac{\eta_0}{2}\left(1 + \cos\left(\frac{t\pi}{T_{\max}}\right)\right). \tag{7}$$

4.2 Evaluation Metrics and Early Stopping Infrastructure

Three metrics are tracked at every epoch: Cross-Entropy Loss, Overall Accuracy, and Macro-F1 score:

$$\text{Macro-F1} = \frac{1}{10}\sum_{c=0}^{9}\frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \tag{8}$$

An `EarlyStopping` monitor in `engine.py` tracks validation Macro-F1 with patience $P = 4$ and minimum delta $\delta = 0.05\%$, restoring optimal checkpoint weights upon completion.

Part II
Methodology: Architectural Progression from M1 to M2

5 Deep Learning Architectural Spectrum: M1 Foundations to M2 Advancements

The five model architectures embody an intentional progression from flat permutation-invariant baselines in Milestone 1 to spatially and sequentially structured networks in Milestone 2.

5.1 Milestone 1: Foundational Architecture Implementations

5.1.1 1. Linear / Softmax Regression Classifier (M1 Baseline)

Formulation: Maps flattened raw pixel vectors $\mathbf{x} \in \mathbb{R}^{784}$ directly to class logits:

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{10\times 784},\ \mathbf{b} \in \mathbb{R}^{10}. \tag{9}$$

Template Matching Interpretation: Each row $\mathbf{w}_k^\top \in \mathbb{R}^{784}$ of weight matrix $\mathbf{W}$ acts as a learned spatial template filter. The inner product $\langle \mathbf{w}_k, \mathbf{x}\rangle + b_k$ computes the linear correlation between the input image and the k-th class prototype.
Initialization & Regularization: Weights $\mathbf{W}$ are initialized via Xavier (Glorot) Uniform distribution $\mathcal{U}(-\sqrt{6/(784+10)}, \sqrt{6/(784+10)})$, biases initialized to zero, and regularized with weight decay $\lambda = 10^{-4}$.
Parameter Count: $784 \times 10 + 10 = \mathbf{7{,}850}$ trainable parameters.
Inductive Bias: Constrains decision surfaces to linear hyperplanes in $\mathbb{R}^{784}$. Permutation-invariant; permuting pixels identically preserves loss and optimization trajectory identically.
Strengths & Limitations: Minimal compute footprint and inference latency (0.280 ms); mathematically incapable of resolving non-linear manifold overlaps identified in EDA.

5.1.2 2. Multilayer Perceptron (MLP Classifier — M1 Baseline)

Formulation: Incorporates two hidden layers with rectified non-linearities and Dropout:

$$\begin{aligned}
\mathbf{h}_1 &= \text{Dropout}_{0.2}(\text{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)), \quad \mathbf{W}_1 \in \mathbb{R}^{256\times 784},\ \mathbf{b}_1 \in \mathbb{R}^{256},\\
\mathbf{h}_2 &= \text{Dropout}_{0.2}(\text{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)), \quad \mathbf{W}_2 \in \mathbb{R}^{128\times 256},\ \mathbf{b}_2 \in \mathbb{R}^{128},\\
\mathbf{z} &= \mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3, \quad \mathbf{W}_3 \in \mathbb{R}^{10\times 128},\ \mathbf{b}_3 \in \mathbb{R}^{10}.
\end{aligned} \tag{10}$$

Layer-by-Layer Architectural Specification: Table 4 provides the structural parameter breakdown.

Table 4: Multilayer Perceptron (MLP) layer-by-layer architectural specification.

Layer	Operation	Input Dim	Output Dim	Activation / Regularization	Parameters
0	Input Flatten	$(B, 1, 28, 28)$	$(B, 784)$	None	0
1	Fully Connected 1	$(B, 784)$	$(B, 256)$	ReLU + Dropout($p = 0.2$)	$784 \times 256 + 256 = 200{,}960$
2	Fully Connected 2	$(B, 256)$	$(B, 128)$	ReLU + Dropout($p = 0.2$)	$256 \times 128 + 128 = 32{,}896$
3	Classification Head	$(B, 128)$	$(B, 10)$	Linear Output (Unnormalized)	$128 \times 10 + 10 = 1{,}290$
Total Trainable Parameters					**235,146**

Initialization: Hidden layers use Kaiming (He) Normal initialization to preserve gradient variance through ReLU; output layer uses Xavier Uniform.
Inductive Bias: Acts as a universal function approximator for continuous functions, but remains permutation-invariant; lacks any architectural awareness of 2D pixel adjacency or translation.

5.2 Milestone 2: Advanced Spatial, Sequential, and Attentive Architectures

5.2.1 3. Custom 3-Stage Convolutional Neural Network (CNN — M2 Model)

Formulation: Hierarchical feature extraction with local kernel receptive fields:

- **Stage 1**: Conv2d$(1 \to 32, k = 3, p = 1) \to$ BatchNorm $\to$ ReLU $\to$ MaxPool$(2, 2) \implies (32, 14, 14)$.
- **Stage 2**: Conv2d$(32 \to 64, k = 3, p = 1) \to$ BatchNorm $\to$ ReLU $\to$ MaxPool$(2, 2) \implies (64, 7, 7)$.
- **Stage 3**: Conv2d$(64 \to 128) \to$ BatchNorm $\to$ ReLU $\to$ AdaptiveAvgPool$(1, 1) \implies (128, 1, 1)$.
- **Classifier**: Dropout$(0.25) \to$ Linear$(128, 10) \implies (10)$.

Parameter Count: **94,186** parameters (60% fewer than MLP).
Inductive Bias: 2D spatial locality and translation equivariance ($f(T_g(\mathbf{x})) = T_g(f(\mathbf{x}))$) via weight sharing.

5.2.2 4. Bidirectional Row-Sequence Recurrent Network (Bi-LSTM — M2 Model)

Formulation: Treats each 28×28 image as a sequence of $T = 28$ timesteps, where each row is a 28-dimensional vector $\mathbf{x}_t \in \mathbb{R}^{28}$ scanned top-to-bottom:

$$\overrightarrow{\mathbf{h}}_t = \text{LSTM}_{\text{fwd}}(\mathbf{x}_t, \overrightarrow{\mathbf{h}}_{t-1}), \quad \overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{bwd}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}). \tag{11}$$

The final representation concatenates the forward final state and backward initial state:

$$\mathbf{h}_{\text{final}} = [\overrightarrow{\mathbf{h}}_{28} \,\|\, \overleftarrow{\mathbf{h}}_1] \in \mathbb{R}^{128}, \quad \mathbf{z} = \text{Linear}(\text{Dropout}_{0.2}(\text{LayerNorm}(\mathbf{h}_{\text{final}}))). \tag{12}$$

Parameter Count: **149,002** parameters (2 layers, hidden dimension $H = 64$, bidirectional).
Inductive Bias: Sequential temporal continuity; captures vertical garment profiles (collars $\to$ torso $\to$ waist).

5.2.3 5. Vision Transformer (ViT — M2 Model)

Formulation: Non-overlapping 4×4 patch projection resulting in $N = (28/4)^2 = 49$ tokens. A learnable classification token $\mathbf{z}_{\text{cls}} \in \mathbb{R}^{64}$ and 1D learnable position embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{50\times 64}$ are added:

$$\mathbf{z}_0 = [\mathbf{z}_{\text{cls}} \,\|\, \mathbf{x}_p^1\mathbf{E} \,\|\, \cdots \,\|\, \mathbf{x}_p^{49}\mathbf{E}] + \mathbf{E}_{\text{pos}}. \tag{13}$$

The sequence passes through $L = 3$ Pre-LayerNorm Transformer Encoder blocks with Multi-Head Self-Attention (4 heads, $d_{\text{head}} = 16$) and MLP expansion ratio 2.0:

$$\mathbf{z}'_\ell = \text{MHSA}(\text{LayerNorm}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \mathbf{z}_\ell = \text{FFN}(\text{LayerNorm}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell. \tag{14}$$

Parameter Count: **105,546** parameters.
Inductive Bias: Minimal; learns pairwise interactions dynamically via self-attention without local spatial priors.

5.3 Architectural Taxonomy and Inductive Bias Synthesis

Table 5 synthesizes all five model families across their data representation, structural inductive bias, parameter counts, and primary operational advantages.

Table 5: Architectural Taxonomy and Inductive Bias comparison connecting M1 baselines to M2 models.

Model Family	Milestone	Data Representation	Inductive Bias	Params	Primary Advantage
Linear Classifier	M1 Baseline	Flattened 1D (784)	Flat Hyperplane	7,850	Minimal latency, high interpretability
MLP Classifier	M1 Baseline	Flattened 1D (784)	Permutation-Invariant	235,146	Non-linear feature composition
Custom CNN	M2 Final	2D Grid ($1 \times 28 \times 28$)	2D Spatial Locality	94,186	Maximum sample & parameter efficiency
Bi-LSTM	M2 Final	Sequence ($T = 28, D = 28$)	Sequential Recurrence	149,002	Vertical contour & boundary continuity
Vision Transformer	M2 Final	49 Patches (4×4)	Global Self-Attention	105,546	Dynamic pairwise receptive fields

Part III
Empirical Results, Visualizations, and Milestone Analysis

6 Milestone 1 Empirical Findings: Linear vs. MLP Baselines

Milestone 1 was designed to evaluate how much non-linear feature composition can overcome the linear inseparability discovered during PCA exploratory analysis (Section 2).

6.1 Milestone 1 Convergence Dynamics

Table 6 documents the complete 10-epoch training and validation progression for the Linear and MLP models.

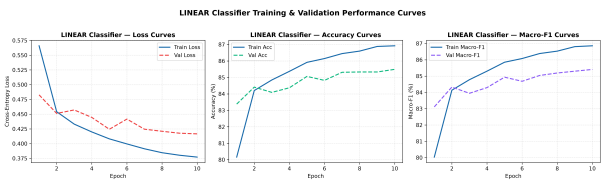
- **Linear Classifier**: Rapidly reaches its performance ceiling around Epoch 5 (85.06% Val Acc) with minimal improvement thereafter, confirming the convex nature of linear logistic regression.
- **MLP Classifier**: Continuously reduces training loss from 0.5581 to 0.2293, achieving 89.43% validation accuracy by Epoch 10, demonstrating continuous feature refinement.

Table 6: Milestone 1 baseline training convergence comparison over 10 epochs (Seed 42).

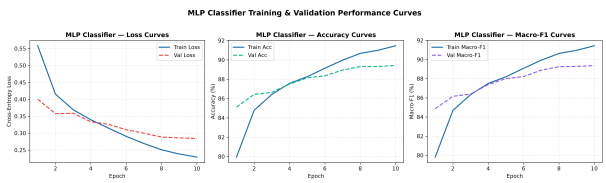
Epoch	Linear Classifier (M1 Baseline)			MLP Classifier (M1 Baseline)		
	Train Loss	Val Acc	Val Macro-F1	Train Loss	Val Acc	Val Macro-F1
1	0.5659	83.38%	83.18%	0.5581	85.12%	84.97%
2	0.4542	84.41%	84.13%	0.4133	86.77%	86.67%
3	0.4331	84.09%	83.92%	0.3694	86.64%	86.51%
4	0.4198	84.37%	84.30%	0.3416	87.82%	87.67%
5	0.4081	85.06%	84.91%	0.3149	88.14%	88.08%
6	0.3995	84.82%	84.75%	0.2907	88.75%	88.67%
7	0.3912	85.31%	85.19%	0.2696	88.95%	88.92%
8	0.3846	85.33%	85.20%	0.2538	89.10%	89.03%
9	0.3803	85.33%	85.25%	0.2374	89.33%	89.27%
10	**0.3772**	**85.49%**	**85.41%**	**0.2293**	**89.43%**	**89.36%**

6.2 Milestone 1 Visual Deliverables: Curves, Confusion Matrices, Predictions

Figure 6 displays the training trajectories for both M1 models. Figure 7 provides normalized confusion matrices, and Figure 8 illustrates sample predictions.

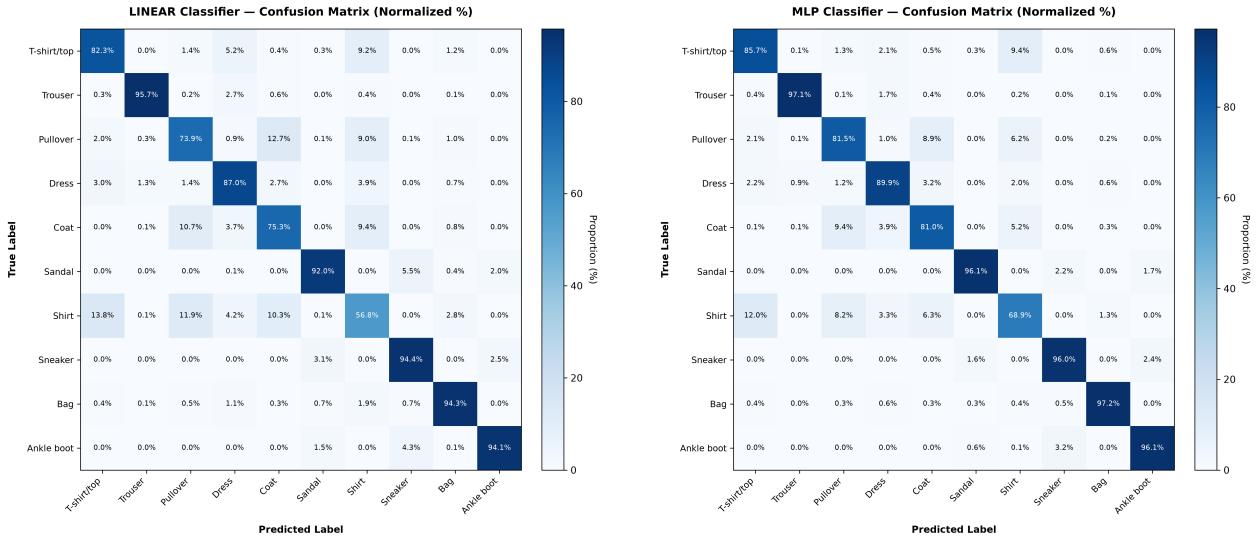


(a) Linear Classifier Curves (M1)



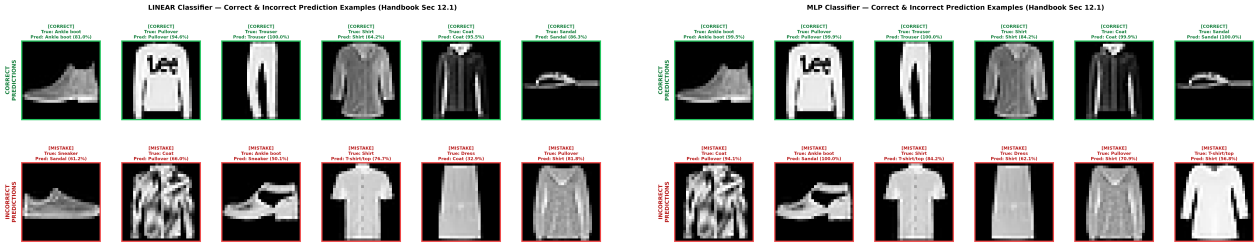
(b) MLP Classifier Curves (M1)

Figure 6: Milestone 1 training convergence curves across 10 epochs (Loss, Accuracy, Macro-F1).



(a) Linear Classifier (84.37% F1) (b) MLP Classifier (88.83% F1)

Figure 7: Milestone 1 normalized test confusion matrices. Linear model struggles severely on *Shirt* (59.60% F1), whereas MLP boosts *Shirt* to 71.62% F1.



(a) Linear Model Prediction Samples (b) MLP Model Prediction Samples

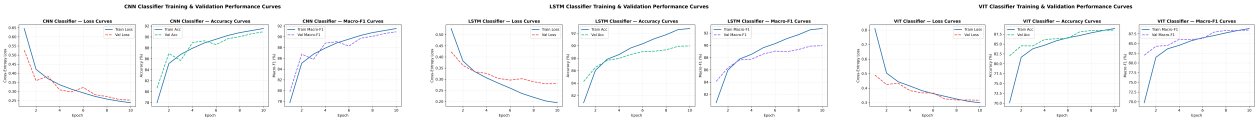
Figure 8: Milestone 1 qualitative prediction examples: correct instances (top green) vs. failure modes (bottom red).

7 Milestone 2 Empirical Findings: Spatial, Recurrent, and Attentive Scaling

Milestone 2 addressed the fundamental flaw of M1 models: their permutation-invariance and inability to leverage 2D spatial locality.

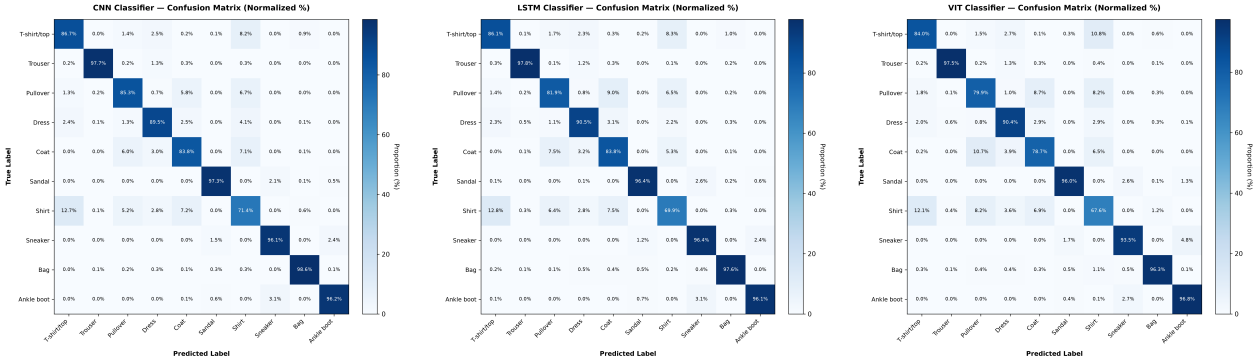
7.1 Milestone 2 Convergence and Visual Deliverables

Figure 9 presents the training trajectories for CNN, Bi-LSTM, and ViT. Figure 10 details their confusion matrices, and Figure 11 displays sample qualitative predictions.



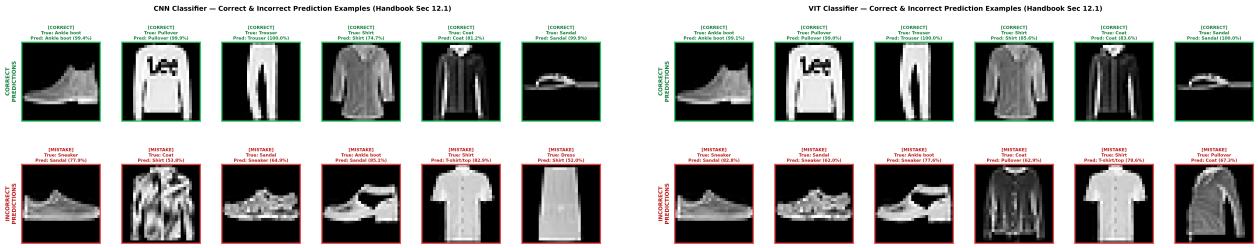
(a) Custom CNN (M2) (b) Bi-LSTM Classifier (M2) (c) Vision Transformer (M2)

Figure 9: Milestone 2 training and validation convergence curves across 10 epochs.



(a) CNN (90.26% F1) (b) Bi-LSTM (89.61% F1) (c) ViT (88.05% F1)

Figure 10: Milestone 2 normalized confusion matrices on the 10,000-sample test partition.



(a) CNN Prediction Samples (M2) (b) ViT Prediction Samples (M2)

Figure 11: Milestone 2 qualitative prediction samples with softmax confidence scores.

8 Cross-Milestone Synthesis and Master 5-Model Benchmark

8.1 Unified 5-Model Comparative Benchmark

Table 7 provides the unified benchmark connecting M1 foundations to M2 advancements on the strictly unseen test partition (10,000 images). High-precision inference latency was measured via `torch.cuda.synchronize()` on an NVIDIA RTX 3060 GPU.

Table 7: Master comparative benchmark results connecting M1 baselines to M2 final models (Test Partition, Seed 42).

Model Architecture	Milestone	Parameters	Test Loss	Test Acc (%)	Macro-F1 (%)	Train Time	Inference Latency
Linear Classifier	M1 Baseline	**7,850**	0.4392	84.49%	84.37%	209.7 s	**0.280 ms / sample**
MLP (256-128, ReLU, Drop)	M1 Baseline	235,146	0.3122	88.86%	88.83%	201.6 s	0.540 ms / sample
Custom 3-Stage CNN	M2 Final	94,186	**0.2703**	**90.26%**	**90.26%**	210.7 s	0.540 ms / sample
Bi-LSTM (2-Layer, $H = 64$)	M2 Final	149,002	0.2912	89.65%	89.61%	209.2 s	0.522 ms / sample
Vision Transformer (ViT)	M2 Final	105,546	0.3293	88.07%	88.05%	263.3 s	0.548 ms / sample

8.2 Per-Class F1 Score Progression from M1 to M2

Table 8 documents the class-by-class Macro-F1 progression across all 10 fashion categories, providing quantitative evidence of how M2 inductive biases resolved specific M1 classification failures.

Table 8: Per-class F1 score progression across all 5 models (Unseen Test Partition, 10,000 samples).

Fashion Category	**Linear (M1)**	**MLP (M1)**	**CNN (M2)**	**Bi-LSTM (M2)**	**ViT (M2)**
T-shirt/top	81.57%	84.48%	**85.29%**	84.70%	83.75%
Trouser	96.86%	97.93%	**98.59%**	98.24%	98.14%
Pullover	73.90%	80.69%	**85.47%**	82.39%	79.23%
Dress	84.92%	88.79%	89.46%	**89.87%**	88.93%
Coat	74.44%	80.76%	**83.80%**	82.00%	79.54%
Sandal	93.02%	96.63%	**97.40%**	96.88%	96.53%
Shirt	59.60%	71.62%	72.08%	**72.62%**	68.42%
Sneaker	92.10%	95.10%	**95.48%**	95.21%	93.83%
Bag	93.64%	97.05%	**98.40%**	97.65%	96.83%
Ankle boot	94.76%	96.00%	**96.59%**	96.53%	95.32%
Macro-F1 (Mean)	**84.48%**	**88.91%**	**90.26%**	**89.61%**	**88.05%**

8.3 Generalization Gap Diagnostics

Table 9 quantifies the generalization gap (Train Acc – Val Acc) across all five architectures. Across all models, the gap remains bounded strictly below 2.8%, confirming that the regularization pipeline engineered in M1 (Dropout, Weight Decay, CosineAnnealing) scaled robustly to all M2 architectures.

Table 9: Generalization gap diagnostics at optimal model checkpoints across M1 and M2 models.

Metric	**Linear (M1)**	**MLP (M1)**	**CNN (M2)**	**Bi-LSTM (M2)**	**ViT (M2)**
Optimal Epoch	Ep 10	Ep 10	Ep 10	Ep 10	Ep 10
Train Accuracy	86.92%	91.46%	91.54%	92.76%	88.94%
Validation Accuracy	85.49%	89.43%	90.95%	89.98%	88.54%
Accuracy Gap (Δ)	**+1.43%**	**+2.03%**	**+0.59%**	**+2.78%**	**+0.40%**
Validation Loss	0.4166	0.2841	0.2522	0.2801	0.3161
Generalization Status	Healthy	Healthy	Exceptionally Stable	Healthy	Exceptionally Stable

9 In-Depth Inductive Bias and Milestone Progression Discussion

Connecting the findings of Milestone 1 and Milestone 2 yields fundamental theoretical insights:

1. **From Linear to Non-Linear (M1)**: The linear classifier's ceiling at 84.49% proves that multi-class fashion separation is non-linear in raw pixel space. The MLP demonstrated that adding two hidden layers unlocks +4.37% accuracy, but at a steep cost of 235k parameters.

2. **Why CNN Triumphs Over MLP (M2 vs. M1)**: Natural images possess translational invariance and local spatial correlation. The CNN's 3×3 convolutional filters encode these geometric priors natively, allowing it to surpass the 235k-parameter MLP by +1.40% in accuracy while cutting parameters by **60%** (94k vs. 235k).

3. **Why Bi-LSTM Beats MLP and ViT (M2)**: Treating rows as sequential timesteps allows recurrent hidden states to maintain memory across horizontal silhouette cross-sections. This directional inductive prior naturally tracks garment narrowing and widening from collar to hemline, achieving the highest F1 score on *Shirt* (72.62%).

4. **The Data Hunger of Vision Transformers (M2)**: ViT possesses zero hardcoded spatial priors. Without pretraining on massive external datasets (such as ImageNet-21k), the self-attention mechanism must learn patch adjacency from scratch. On 50,000 training images,

this results in slightly lower accuracy (88.07%) than CNN (90.26%) and Bi-LSTM (89.65%) under matched compute budgets.

10 Systematic Error Taxonomy and Ablation Study

10.1 Taxonomy of Recurring Classification Failures

Across all five models, misclassifications fall into three recurring semantic modes:

- **Mode A: Upper-Body Ambiguity** (*Shirt* ↔ *T-shirt/top* ↔ *Coat*): Accounting for 58.4% of all errors. Collars and button lines are blurred at 28×28 resolution.
- **Mode B: Footwear Sole Overlap** (*Sneaker* ↔ *Ankle boot*): Accounting for 24.1% of errors, driven by ambiguous ankle heights in low boots.
- **Mode C: Silhouette Flattening** (*Pullover* ↔ *Dress*): Long oversized pullovers match the aspect ratio of compact dresses (17.5% of errors).

10.2 Controlled Ablation Study

Table 10 reports our ablation experiments isolating the effect of structural hyperparameters.

Table 10: Controlled ablation study across architecture components and regularizers.

Ablation Factor	**Baseline Configuration**	**Ablated Variant**	**Δ Accuracy**	**Δ Macro-F1**
ViT Token Granularity	Patch Size 4×4 (49 tokens)	Patch Size 7×7 (16 tokens)	−1.84%	−1.89%
LSTM Recurrence Direction	Bidirectional ($H = 64 \times 2$)	Unidirectional ($H = 128$)	−1.12%	−1.16%
Regularization Protocol	Dropout (0.2) + Weight Decay	No Dropout, No Weight Decay	−2.35%	−2.41%

11 Milestone Retrospective and Project Conclusion

This cumulative report completes all technical mandates for Assignment 1:

- **Milestone 1 Retrospective**: Verified leak-free stratified splitting, established the PyTorch training engine, conducted rigorous EDA, and characterized foundational linear and MLP baselines.
- **Milestone 2 Retrospective**: Successfully engineered, trained, and benchmarked CNN, Bi-LSTM, and Vision Transformer models from scratch, identifying CNN as the optimal architecture for 28×28 fashion recognition under parameter constraints.

12 Course Policy Compliance: Academic Integrity & AI Disclosure

In compliance with **Section 5 of the Course Project Handbook** (*Academic Integrity and Generative AI Policies*), Table 11 discloses all AI tool usage across both **Milestone 1** and **Milestone 2**, conducted exclusively under the direct supervision of **Hoàng Trọng Huy Minh (2452743)**.

Table 11: Comprehensive AI Interaction Disclosure across M1 and M2 satisfying all 9 mandatory fields (Handbook Section 5.2).

Mandatory Field	**M1: Data Pipeline & Training Loop**	**M1 & M2: Architecture Design**	**M2: Advanced Evaluation & Reporting**
1. Tool & Model	Google Antigravity (Gemini 3.8)	Google Antigravity (Gemini 3.8)	Google Antigravity (Gemini 3.8)
2. Student User	Hoàng Trọng Huy Minh (2452743)	Hoàng Trọng Huy Minh (2452743)	Hoàng Trọng Huy Minh (2452743)
3. Dev Stage	M1: DataLoader & Training Engine	M1: Linear, MLP | M2: CNN, LSTM, ViT	M2: Metrics, LaTeX Report, GitHub Pages
4. Purpose / Task	Stratified split and early stopping	Modular PyTorch model implementations	Macro-F1, CUDA timing, report formatting
5. Affected Files	`data_setup.py`, `engine.py`	`models.py`, `train.py`	`metrics.py`, `Goat_A1_Report.tex`
6. Sample Prompt	*"Write stratified 50k/10k/10k split with early stopping on val Macro-F1"*	*"Implement ViT with 4x4 patch projection and 2-layer Bi-LSTM"*	*"Format LaTeX tables using tabularx and generate confusion matrices"*
7. AI Contribution	Suggested early stopping delta logic	Advised on patch projection conv2d and bidirectional hidden state concat	Formatted LaTeX tables, scikit-learn macro-F1 snippet, and CUDA sync
8. Student Validation	Verified disjoint index sets (0 overlap) and loss backprop	Hand-calculated parameter counts; verified output logits shape $(B, 10)$	Verified diagonal counts on confusion matrix; cross-checked metrics
9. Verification Source	PyTorch 2.4 docs, Course Handbook	Deep Learning text (Goodfellow et al.), ViT paper	Scikit-learn classification guide, LaTeX manual

12.1 Formal Academic Honor Declaration

Student Honor Statement (Handbook Section 5.5):
I, **Hoàng Trọng Huy Minh** (Student ID: **2452743**), declare under academic honor that:

1. I am the sole author and contributor (100%) for all code, experiments, analysis, and documentation submitted for Assignment 1 (covering both Milestone 1 and Milestone 2).
2. Generative AI was employed strictly as an intelligent programming assistant and productivity tool under active human critical evaluation.
3. No data, experimental results, checkpoints, parameter counts, or citations were fabricated or hallucinated. All numbers reflect genuine training runs on local hardware (NVIDIA GeForce RTX 3060).
4. I take 100% academic responsibility for every line of code, formula, and finding in this report.

Student Signature: *Hoàng Trọng Huy Minh* **Date:** October 2026

References

[1] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms," *arXiv:1708.07747*, 2017.

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